Inference Time Adaptation of a Surgical Video Language Model

Valmik Nahata

University of California, San Diego, CA, USA
vnahata@ucsd.edu

Abstract. Benchmarks for surgical video language models score submissions with an automatic evaluator and keep the test labels private. If the code around a model writes answers in a format the evaluator does not expect, correct answers are scored as wrong, and the leaderboard gives no sign of it. We met this problem while adapting a released model to MedVidBench without changing its weights. To find such faults, we list the answer format the evaluator accepts for each task, pass ground truth through our own code to check that it scores perfectly (the *answer key test*), and compare statistics of our predictions with those of the public data (the *output audit*). These steps revealed boxes written with swapped coordinates and a decoding setting that emptied most predicted boxes. Fixing both, and adding frame timestamps to the prompt, gave the largest improvements in our controlled comparisons.

Keywords: Surgical video understanding · Vision language models · Inference time adaptation · Evaluation

1 Introduction

MedVidBench [37] evaluates surgical video language models on eight tasks: Critical View of Safety assessment (CVS), next action prediction (NAP), skill assessment (SA), spatio temporal grounding (STG), temporal action grounding (TAG), dense video captioning (DVC), video summarization (VS) and region captioning (RC). A single model therefore has to produce ratings, labels from closed sets, time spans in seconds, bounding boxes keyed by timestamp, and captions that a language model judge grades [22, 48]. The test labels are private, and a submission to the leaderboard returns one number per metric. Figure 1 shows one sample of each task.

A low score is therefore difficult to interpret. The model may lack the skill, or the code around it may have lost an answer that it got right. This happened in our own work. Our first pipeline and the evaluator disagreed on the order of box coordinates, so that correct boxes were scored as misses. The leaderboard reported a low grounding score, and for some time we attributed it to the model.

We address this problem for a released checkpoint whose weights we do not change. Fine tuning a model of this size was beyond our compute, and a frozen model has the advantage that every change in score can be attributed to the code

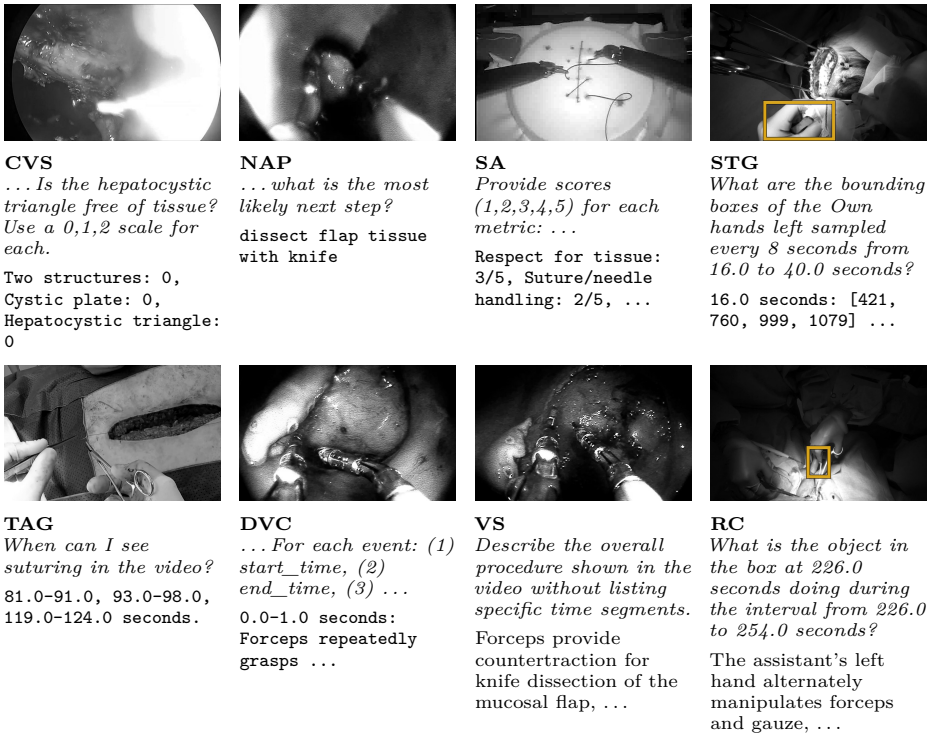


Fig. 1: One sample of the MedVidBench public split [37] (CC BY-NC-SA 4.0) for each task, with its question and its ground truth answer as the evaluator expects it. Gold boxes are annotations. Long questions and answers are cut (. . .).

around it. Our contribution is a procedure for finding where points are lost when the test set is hidden. It consists of a format card and two checks, the answer key test and the output audit, that need only the public split and the source of the evaluator. The procedure led to five changes at inference time, three of which controlled comparisons confirm, and the submission placed second among entrants to the MedVidU challenge at ECCV 2026.

2 Related Work

Benchmark scores depend on how an evaluation is set up. For language models they move with prompt format [33], instruction wording [24] and the order of answer options [47], enough to reorder leaderboards [2], as tokenization once did for BLEU [28]. Evaluation toolkits therefore standardize prompts and answer extraction [5, 10, 43, 46], but for text and multiple choice answers. Error diagnosis tools [1, 6, 15, 31] analyze the model and require labels, whereas our checks ask, without test labels, whether the pipeline reports what the model did.

For grounding, the conventions that matter are coordinate spaces and time units. Models that write boxes as text put the vertical coordinate first [4,9] or the horizontal [42], use location tokens [27], and within one family changed from a normalized range [40] to absolute pixels [3]. Sampled frames carry no timestamps, and video models supply them through training [8, 17–19, 30, 41]. Timestamp prompting changes only the prompt.

Surgical vision language models have grown from contrastive pretraining [44] and question answering [34] into multitask assistants [20, 32, 37, 45]. Spatial and temporal tasks remain difficult for them [29], and some answers do not depend on the image [7, 36].

3 Method

3.1 Setting

We use the checkpoint released with the benchmark [37], a Qwen2.5-VL-7B [3] backbone that was trained further on the benchmark's training data, first with supervised fine tuning and then with group relative policy optimization [35]. The benchmark scores 6,245 test samples whose videos come from eight public datasets [11, 12, 14, 16, 25, 26, 38, 39]. The benchmark also publishes a labelled split, from which we hold out 1,640 samples for our checks, and the source code of its evaluator.

Table 1 lists, for every task, the text the evaluator parses and the rule it scores by, read from the evaluation code that the benchmark publishes with its leaderboard (version of July 2026).

3.2 Answer Key Test

The answer key test is run before any inference. We take ground truth answers from the held out samples, write each one in the format that our prompt asks the model to produce, and send it through our own parsing and post processing into the official evaluator. Ground truth must score as correct, so any shortfall is caused by formatting and not by the model. The test needs neither the model nor test labels.

An earlier pipeline of ours failed this test. Ground truth boxes returned 0.120 mIoU on STG. The cause was the order of the coordinates. We had requested boxes as [ymin, xmin, ymax, xmax], whereas the evaluator reads [x1, y1, x2, y2]. That pipeline had already scored 0.150 mIoU on the leaderboard. After the order was corrected, the test passed.

3.3 Output Audit

The output audit covers what the model writes, by comparing simple statistics of our predictions with those of ground truth in the public split (Tab. 2). The audit found two faults that the aggregate score conceals. Under one of our decode

Table 1: Format card for MedVidBench: the answer text that the official evaluator parses for each task and the rule it scores by.

Task	Answer the evaluator parses	Scoring rule
CVS	`Two structures: a, Cystic plate: b, Hepatocystic triangle: c`	Exact match per criterion, averaged; a missing criterion is skipped.
NAP	The label as a string	Exact match; other text goes to the nearest label by sentence embedding.
SA	`Aspect: n/5` for each aspect, separated by commas	Exact match per aspect, averaged; names must match ground truth.
STG	`t seconds: [x1, y1, x2, y2]` per timestamp, in original pixels	Mean IoU over ground truth timestamps. A missing one reuses the previous box. Extra ones are ignored.
TAG	Spans in seconds written as `start-end`	Mean IoU of ground truth spans matched to unused predictions above the threshold.
DVC	`start-end seconds: caption` per segment	Temporal F1 over IoU 0.3, 0.5 and 0.7. The judge grades only matched segments.
VS, RC	Free text	GPT-4.1 judge, five aspects from 1 to 5, averaged.

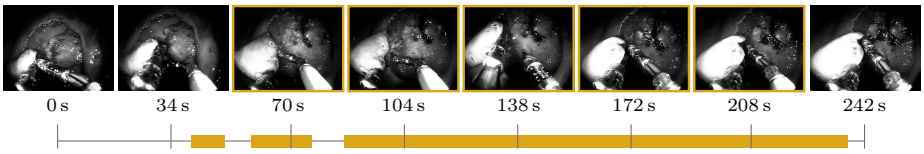
Table 2: Output audit: statistics of our test predictions and of ground truth in the public split.

Statistic	Predictions	Ground truth
STG timestamps answered per sample	4.75	4.66
TAG spans per sample	1.32	1.99
TAG samples with several spans	17.0%	39.5%
RC caption length, median words	24	22
VS caption length, median words	38	27
DVC caption length, median words	104	139

settings, most predicted boxes had zero area, which we trace to the coordinate space in Sec. 4. The second fault is a tendency to under segment in TAG, where the model returns a single span for questions whose ground truth often contains several.

3.4 Changes at Inference Time

The two checks led to five changes, two of which concern boxes. The answer key test settled the order of the coordinates but left open the space in which the model writes them, and the backbone family suggests two candidates: coordinates normalized to a fixed range [40] and absolute pixels of the resized input [3]. We built a decoder for each, which we call `norm` and `abs`, and compared both with *native decoding*, which keeps the coordinates exactly as the model writes them.



Question. What is the time span of dissect flap tissue with knife?

Added by timestamp prompting. Temporal reference: the clip is 242.0 seconds long. The provided frames are at these timestamps: 0.0s, 34.0s, 70.0s, 104.0s, 138.0s, 172.0s, 208.0s, 242.0s. Report all times in seconds within [0, 242.0].

Fig. 2: Timestamp prompting on a sample of the MedVidBench public split [37] (CC BY-NC-SA 4.0), shown with eight sampled frames. The shaded sentence is what our pipeline adds to the question. Gold marks the ground truth answer and the frames inside it.

Table 3: Controlled comparisons on the leaderboard. Each row varies one setting with everything else fixed.

Change	Metric	Alternatives	Adopted
Native decoding	STG mIoU	`abs` 0.003, `norm` 0.076	native **0.135**
Timestamp prompting	TAG mIoU at 0.3	without 0.074	with **0.222**
Per task resolution	CVS accuracy	reduced 0.733	full **0.890**

The second change follows from the way STG is scored. Questions name an exact window and stride, such as “every 8 seconds from 66 to 78”, and the evaluator reads only the timestamps of ground truth (Tab. 1). With *query focused coverage* we therefore answer every timestamp that the question implies.

The third change concerns time. Frames reach the model with no indication of when they occur, so with *timestamp prompting* we add one sentence to the prompt that states the timestamp of every sampled frame and the duration of the clip (Fig. 2).

The remaining two changes concern the visual budget and the captions. *Per task resolution* keeps the full input resolution for tasks that depend on small structures and lowers it only for tasks that are dominated by language. *Style calibrated captioning* rewrites the caption prompts so that they follow the five aspects of the judge’s rubric, and it adds one exemplar per task.

4 Experiments

Table 3 reports the three changes that we could test with leaderboard submissions differing in a single setting. Native decoding outperforms both alternatives by a wide margin. The audit explains the weakest setting, since 73.5% of the boxes predicted under `abs` have zero area. The checkpoint was trained with reinforcement learning on the benchmark’s own tasks, and it appears to have learned to write boxes in the evaluator’s convention already, so that both of our

Table 4: MedVidBench test results of the submitted pipeline.

CVS	NAP	SA	STG	TAG@0.3	TAG@0.5	DVC F1	DVC	VS	RC
.895	.382	.247	.135	.162	.106	.217	3.59	4.19	3.64

decoders moved its output away from the correct space. The format a model writes is set by its last stage of training and should be verified before it is corrected.

Timestamp prompting was our largest single gain and required no other change. Without timestamps, a model trained for temporal grounding reaches only 0.074 mIoU on TAG at 0.3. Lowering the resolution left the tasks dominated by language largely unchanged, but it cost CVS about a sixth of its accuracy, because CVS depends on small anatomical structures.

Query focused coverage and style calibrated captioning have no controlled test, so we attribute no gain to them. The audit (Tab. 2) is their only evidence.

Table 4 lists the test scores of the submitted pipeline. Three submissions of identical predictions returned judge scores within ± 0.003 of each other. We do not compare with the scores published for the checkpoint, which come from the authors' own inference code.

5 Discussion and Conclusion

The procedure applies to any benchmark that publishes its evaluator and a labelled split. One writes down the format the evaluator parses for each task and how it treats missing or additional items, confirms that ground truth sent through the output code scores as correct, and compares simple statistics of the predictions with the labelled split.

Query focused coverage and style calibrated captioning exploit rules of this evaluator and judge, whereas we expect the other three, which address the model and the task, to transfer. Each comparison in Tab. 3 is a single run. Interrupted runs were resumed by sample and not by configuration, so the submitted TAG predictions may include stale ones, which may explain why the submitted TAG@0.3 score is below the controlled result.

For future work, benchmarks could release a format card and an answer key test with their evaluator, in the spirit of datasheets and model cards [13, 23]. Evaluators could flag a swapped axis order or a normalized coordinate space instead of returning a low score, as learned extractors do for text [43]. An input side check, comparing frame sampling and resolution with those used in training, would cover faults our checks cannot see. Audit statistics could be related to leaderboard scores across many submissions to estimate performance without labels, and reports should state the inference pipeline next to every score [5, 21]. For this model, decoding several spans per question could raise recall in TAG.

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